Neurocomputing 634 (2025) 129878

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

MSSFN: A multi-scale sequence fusion network for CT-based diagnosis of pulmonary complications☆

Hongfu Zeng [a,1], Xinyu Li [b,1], Haipeng Xu [a], Keyi Yu [c], Huihua Hu [a], Peishu Wu [b], Nianyin Zeng [b,*]

[a] The Department of Thoracic Oncology, Clinical Oncology School of Fujian Medical University, Fujian Cancer Hospital, Fujian 350014, China
[b] The School of Aerospace Engineering, Xiamen University, Fujian 361102, China
[c] National Institute for Data Science in Health and Medicine, Xiamen University, Fujian 361102, China

ARTICLE INFO

Communicated by R. Yang

Keywords:
Pulmonary complications
Computed tomography
Transfer learning
Multi-scale feature fusion

ABSTRACT

Pulmonary complications exhibit similar symptoms, such as ground-glass opacities and patchy consolidations, posing significant challenges for manual diagnosis. In this paper, a novel Multi-Scale Sequence Fusion Network (MSSFN) designed for a small-scale Computed Tomography (CT) database is proposed for computer-aided identification of pulmonary complications. The MSSFN first applies a transfer learning mechanism that utilizes the convolutional neural network weights pre-trained on a large dataset, enabling optimization of both feature extraction efficiency and model accuracy. Next, the multi-scale feature fusion module uses different types of convolutional layers in parallel to extract and integrate high-level features, enabling the network to capture complex lung lesion characteristics from multiple perspectives and enhance information interaction across different layers. Finally, a sequence feature fusion module establishes three-dimensional connections within the CT data, ensuring comprehensive data fusion by effectively integrating spatial features across the sequence. The comprehensive experimental results demonstrate that, under the same experimental conditions, the MSSFN achieves an accuracy of 82.54% for identifying pulmonary complications in CT images, which exhibits superior performance compared to nine similar network structures as well as other four state-of-the-art deep learning models. Consequently, the proposed MSSFN demonstrates practical significance by providing radiologists with reliable tools for accurately distinguishing various pulmonary complications in CT images. It also holds theoretical value by advancing methods for constructing robust features with enhanced representational capabilities.

1. Introduction

Pulmonary Lymphangitic Carcinomatosis (PLC) results from the infiltration and dissemination of tumor cells within the pulmonary lymphatic vessels, often signifying advanced disease [1]. With the ongoing advancements in cancer therapies, clinical management of PLC includes not only conventional radiotherapy, but also immune checkpoint inhibitors as a viable treatment option. However, these therapeutic approaches are frequently associated with complications such as Radiation Pneumonitis (RP) and Checkpoint Inhibitor Pneumonitis (CIP) [2,3]. In order to improve patient prognosis and guide precise treatment of PLC, RP, and CIP, it is crucial to develop timely and accurate diagnosis methods. At present, in imaging-based diagnosis of above pulmonary complications, the manual interpretation of Computed Tomography (CT) images by clinicians is typically limited by subjectivity, inefficiency and lack of standardization.

Particularly, PLC, CIP and RP share common features in CT imaging of pathological states, such as ground-glass opacities, interlobular septal thickening, and consolidation, making it difficult to accurately distinguish among the three types of diseases. Traditional classifiers such as logistic regression, random forest and linear SVM [4–6], rely on handcrafted features and specific database, which cannot fully capture the complex and potential imaging characteristics of PLC, CIP and RP.

☆ This work was supported in part by the Natural Science Foundation of China under Grant 62073271, the Fundamental Research Funds for the Central Universities of China under Grant 20720220076, the Natural Science Foundation for Distinguished Young Scholars of the Fujian Province, China under Grant 2023J06010, the National Science and Technology Major Project, China under Grant J2019-I-0013-0013 and the National Key Research and Development Program of China under Grant 2024YFC3407000.

* Corresponding author.
E-mail address: zny@xmu.edu.cn (N. Zeng).

[1] Co-first authors.

https://doi.org/10.1016/j.neucom.2025.129878
Received 11 January 2025; Received in revised form 10 February 2025; Accepted 25 February 2025
Available online 5 March 2025
0925-2312/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

In recent years, an increasing number of Deep Learning (DL)-based algorithms [7–9] have been developed to achieve the CT image differentiation of multiple pulmonary lesions, including Two-Dimensional (2D) and Three-Dimensional (3D) neural networks. For 2D pulmonary CT diagnostics, classical networks such as ResNet, VGG, and their variants are often employed as the backbone for feature extraction [10–13]. Additionally, inspired by the attention mechanism, many 2D networks incorporate attention modules to enhance the ability of the model to capture image details, as highlighted in Refs. [11,12,14,15].

In addition, a popular trend to enhance the CT-based lesion recognition performance is to utilize the 3D neutral networks, which can take both the spatial continuity and temporal dynamics into consideration so that rich underlying imaging features, such as the texture variations and morphological changes can be sufficiently exploited. By utilizing 3D convolution, networks are better adapted to CT images. To avoid the problem of overfitting, ResNet series networks are widely used, as demonstrated in [16–19]. Additionally, to improve recognition accuracy, many 3D networks first perform preliminary segmentation of pulmonary CT images before classification, as noted in [18,20,21].

However, despite the success of 2D- and 3D-based methods for pulmonary CT diagnosis, there are still the following challenges in accurately differentiating PLC, RP, and CIP. Firstly, lack of a unified CT database of the above three types of pulmonary complications. Secondly, not fully utilizing multi-scale lesion image information. Finally, failure to thoroughly analyze information regarding changes in lesion morphology, size, and density over time.

In this article, aiming at CT-based diagnosis of pulmonary complications, a Multi-Scale Sequence Fusion Network (MSSFN) is proposed, which adopts a consistent preprocessing strategy to standardize the number of slices per patient case across the dataset, and by doing so, ensures uniform input dimensions and consistent data quality, thereby addressing potential biases and variations inherent in the dataset, which facilitates more reliable training and evaluation of the model. Then, the GoogLeNet and Multi-Scale Fusion Module are deployed to sufficiently capture and integrate multi-scale lesion image information, expanding the depth and breadth of lesion analysis. In addition, a Bi-directional Long Short-Term Memory (Bi-LSTM) is applied to take sequential changes in CT images into account, which can effectively track and analyze the progression of lesions over time, thereby promoting a nuanced understanding of disease evolution and enhancing diagnostic accuracy for conditions that manifest changes across temporal dimensions.

The contributions of this paper are summarized as follows:

(1) By collecting 313 real clinical cases, a new CT-based pulmonary complications diagnosis database is constructed, including three types of PLC, CIP and RP.
(2) By applying multi-scale convolution operators with the three-branch structure, a diverse range of lesion features across different scales can be effectively extracted.
(3) Temporal changes in 3D imaging features is organically embedded in MSSFN, where the bi-directional sequential patterns are used to promote diagnostic accuracy.

The remainder of this paper is organized as follows. A review of related work on lung diagnosis based on deep learning is presented in Section 2. The detailed implementation of the MSSFN is elaborated in Section 3. The construction of the dataset and experimental results with discussion are displayed in Section 4. Finally, a conclusion of this paper and future prospects are provided in Section 5.

2. Related works

In order to minimize human factor interference and provide comprehensive and objective clinical diagnosis, Computer Aided Diagnosis (CAD) systems for chest CT image classification have become increasingly popular [22,23], enabling automatic delineation of pathological regions. In this section, both 2D and 3D deep learning methods used in CAD will be sequentially reviewed.

2.1. CT image diagnosis based on 2D deep learning

In 2D deep learning methods, the feature extraction strategies for medical images typically play a crucial role [24,25]. Given the high dimensionality and complex structure of medical data, a Hercules model has been proposed in [14], which has applied hierarchical attention mechanisms and multi-level feature fusion techniques. Consequently, this model has employed Monte Carlo (MC) dropout to quantify uncertainty within the attention mechanism, thereby enhancing its capacity to handle uncertainty in input data.

To more effectively extract local and global features from images, a dual-branch strategy has been proposed in [26]. This deep learning network integrates Transformer modules with VGG19 to comprehensively extract CT features, followed by the design of a bidirectional fusion module that facilitates information exchange between the two modalities, ultimately achieving an accuracy of 96.7% on the COVIDx-CT dataset.

Considering the three-dimensional characteristics of the lesions, a Multi-View Knowledge-Based Collaborative (MV-KBC) deep learning model has been introduced in [27]. This model decomposes 3D lung nodules into nine fixed-angle 2D slices and constructs a Knowledge-Based Collaborative (KBC) sub-model for each viewpoint, utilizing three different types of image blocks to fine-tune three pretrained ResNet-50 networks, which represent the overall appearance, voxel heterogeneity, and shape heterogeneity of the nodules. The outputs of these sub-models are combined through an adaptive weighting scheme learned during backpropagation, enabling the MV-KBC model to be trained end-to-end.

It is noteworthy that the aforementioned methods exhibit high network complexity, which is not conducive to small-scale medical image learning tasks. On the other hand, although sufficient feature extraction is achieved at the 2D level, the temporal and spatial variations in CT images are ignored. As a result, numerous studies at the 3D level have emerged.

2.2. CT image diagnosis based on 3D deep learning

3D CNNs have been widely utilized in medical image analysis due to their ability to capture spatial continuity, as demonstrated in various studies [28,29]. For instance, a 3D deep Leaky Noisy-OR Network for the detection and classification of pulmonary nodules has been proposed in [30]. This network employs a 3D U-Net as its backbone and utilizes a Region Proposal Network (RPN) to predict the bounding boxes of nodules. The top five nodules with the highest confidence are selected, and their cancer probabilities are calculated using a Noisy-OR leakage model. Additionally, a Self-Supervised Transfer Learning based on Domain Adaptation (SSTL-DA) 3D CNN framework has been introduced in [31]. By leveraging a self-supervised subnet, domain adaptation subnet, and a pulmonary nodule classification subnet, this framework effectively utilizes unlabeled CT scan data to enhance the accuracy of distinguishing between benign and malignant nodules.

In [18], different segmentation algorithms have first been applied to segment pulmonary regions in order to improve the signal-to-noise ratio of the models. The efficacy of various 3D-CNN networks, including ResNet-50, ResNet-152, DenseNet-169, DenseNet-201, Resnext-50, Seresnet-152, and Seresnext-50, has been evaluated for the classification detection of COVID-19 CT images, with 3D ResNet-50 showing the best performance.

Apart from these 3D-CNN approaches, combinations of CNNs and Recurrent Neural Networks (RNNs) have also provided effective solutions for three-dimensional medical image processing. In [32], features have been extracted from CT scans and chest X-ray images using a multi-level feature extraction strategy involving Global Image Descriptor (GIST), Scale-Invariant Feature Transform (SIFT), and CNNs. Subsequently, an LSTM network has been employed to capture long-term dependencies, thereby enhancing classification accuracy. This

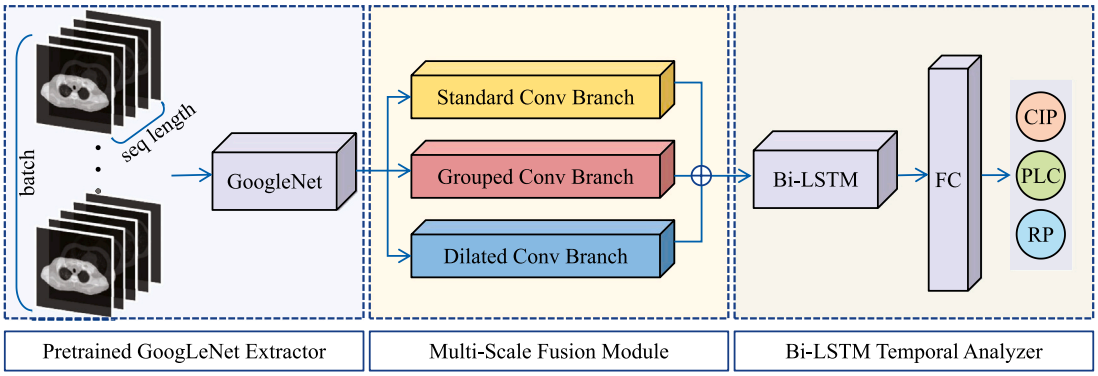


Fig. 1. Framework of the proposed Multi-Scale Sequence Fusion Network (MSSFN).

approach achieved an accuracy of 98.94% on the SARS-CoV-2 CT scan dataset and 83.03% on the SIRM COVID-19 CT scan and chest X-ray datasets. Similarly, feature extraction using pretrained InceptionResNetV2 and ResNet50, followed by integration with LSTM for the classification of benign and malignant subtypes in breast cancer histopathology images, has been proposed in [33].

Based on the above discussions, it is apparent that while extracting 2D lesion information, maintaining spatial continuity of CT sequences remains a challenging task. Therefore, in this study, both multi-scale and spatial information are highlighted. Additionally, specialized data processing methods have been designed for small-scale CT image datasets.

3. Methodology

This section elaborates in detail the implementation specifics of the proposed MSSFN model, which includes a simplified GoogLeNet [34] module, a multi-scale convolution fusion module, and a Bi-LSTM network classifier. The overall network architecture is illustrated in Fig. 1.

3.1. Overall framework of MSSFN

The overall framework of the deep learning-based lung lesion auxiliary diagnostic system designed is shown in Fig. 1. Initially, the preprocessed CT sequences are fed into GoogLeNet, where the learned parameters from large datasets are utilized to efficiently extract preliminary features from the CT images. This process effectively identifies and enhances key features within the CT images while suppressing background noise and irrelevant information, thereby optimizing the feature representation. Subsequently, the multi-scale convolution fusion module is employed to further refine the feature representation. This module includes parallel processing of three types of convolutional branches, which captures multi-level feature details across various scales, thus enhancing the sensitivity of our model to lesions of varying sizes in CT images and boosting the expressiveness of the feature maps. Next, the extracted high-level, rich two-dimensional features are input into a Bi-LSTM for learning spatial relationships, ultimately completing the classification task.

In the following subsection, the three main components mentioned above in the MSSFN will be detailed.

3.2. Feature extraction based on transfer learning

Due to privacy and legal restrictions, the high cost of acquisition, the high heterogeneity and complexity of data, and the difficulty of precise annotation, medical imaging data is often limited in quantity. This poses a particular challenge in the fields of machine learning and artificial intelligence, which typically require large volumes of data to train effective models. To overcome these limitations, transfer learning offers an effective strategy, allowing the use of models pretrained on other large-scale datasets to enhance learning outcomes on smaller or specific-domain datasets [35–37].

In this model, the pretrained GoogLeNet (Inception-v1) is employed as the initial part of feature extraction. GoogLeNet significantly enhances the initial efficiency and accuracy in handling lung CT image classification tasks through its complex parameters trained on the extensive image dataset, ImageNet.

Moreover, to reduce the complexity of the model to better suit the characteristics of lung CT image data, the last five layers of the original GoogLeNet architecture were removed, and the output of the Inception (4e) module is used as the primary result of feature extraction. The working principle of the aforementioned module is represented by the following equation:

$$F_G = GoogLeNet_{17layers}(X_i) \tag{1}$$

where $GoogLeNet_{17layers}$ represents the first 17 layers of GoogLeNet. X_i represents the input to the MSSFN model, initially shaped as $(B \times S, C_0, H_0, W_0)$, and after processing through GoogLeNet, the output feature map F_G has the shape $(B \times S, C_1, H_1, W_1)$. Here, B represents the batch size, S represents the sequence length, and C, H and W represent the depth, height, and width of the feature maps, respectively.

3.3. Integrated multi-scale fusion module (MSFM)

In the field of medical image processing, traditional methods often fail to fully consider the multi-scale and multi-directional characteristics of image data, which limits the performance of models when dealing with heterogeneous lung lesions. To address this issue, an MSFM has been designed to optimize the classification performance of lung CT images. This module includes three parallel convolutional branches, each dedicated to processing image features at different scales. Through this approach, the module integrates multi-scale information, significantly enhancing the ability of MSSFN to analyze complex medical image data.

Specifically, this module further processes the output features F_G obtained from GoogLeNet. Fig. 2 displays the detailed network structure of the MSFM. The standard conv branch includes two serial 3×3 convolution layers, focusing on refining local features in the image to extract detailed information. The grouped conv branch utilizes depthwise separable convolution, which splits the input channels evenly into two parts, each applying a 3×3 grouped convolution. This method not only reduces the number of parameters but also enhances computational efficiency, subsequently integrating features through a 1×1 convolution layer to effectively capture sequential information. The dilated conv branch uses two serial dilated convolutions with dilation rates of 2 and 4, respectively. This design expands the receptive field of the network, allowing the model to capture a broader context

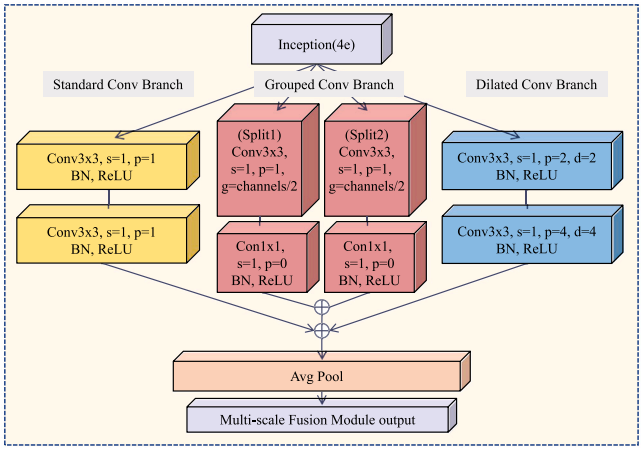


Fig. 2. Architecture of Multi-Scale Fusion Module (MSFM).

without additional computational burden, thus aiding in identifying large-scale structures within the feature maps. The above process can be represented by the following equation.

Standard Conv Branch:

$$F_1 = \sigma(W_1\sigma(W_{1m}F_G)) \tag{2}$$

Grouped Conv Branch:

$$F_2 = \sigma(W_{2a}F_{split1}) \oplus \sigma(W_{2b}F_{split2}) \tag{3}$$

Dilated Conv Branch:

$$F_3 = \sigma(W_3\sigma(W_{3m}F_G)) \tag{4}$$

where F_{split1} and F_{split2} represent the front and back parts of the channel divided equally, σ denotes the $ReLU$ activation function, W_1 and W_{1m} represent the weights of the standard convolution kernels, W_{2a} and W_{2b} represent the weights of the grouped convolution kernels, and W_3 and W_{3m} denote the weights of the dilated convolution kernels.

Following this processing, the outputs from each branch are concatenated along the channel dimension and integrated via adaptive average pooling:

$$F_{fusion} = AvgPool(Concat(F_1, F_2, F_3)) \tag{5}$$

The final shape of F_{fusion} is $(B\times S, C_2, 1, 1)$, ensuring comprehensive extraction and effective fusion of features across different scales. The fundamental principle of multi-scale is the extraction of information at varying scales from the input features, with the objective of enhancing the capability of the model to comprehend complicated structures through the integration of this information. It is noteworthy that, in contrast to conventional multi-scale models, all three convolution kernels utilized in this study are 3×3, a reduction intended to minimize the complexity of the model. By adjusting the dilation rate to obtain a larger field of view, utilizing grouped convolution to integrate global information, and employing standard convolution to focus on local details, the objective of capturing features at varying scales of the input feature map is ultimately realized. Thereby enhancing the accuracy and robustness in classifying lung diseases.

3.4. Temporal dynamics capture with Bi-LSTM

In the first two parts of the network, feature extraction primarily occurs at the horizontal level of each slice. To establish connections between slices longitudinally and to more comprehensively understand spatial variations within lung CT sequences, the final part of the network incorporates a Bi-LSTM module. The introduction of Bi-LSTM is crucial for leveraging its ability to learn forward and backward dependencies in time series data, which is vital for analyzing medical imaging data with a temporal dimension.

Specifically, the output from the MSFM F_{fusion} is first adjusted to fit the shape required for LSTM input. Then, these serialized feature data are processed in the Bi-LSTM module. The Bi-LSTM addresses both forward and reverse sequences simultaneously, allowing it to capture information at each point throughout the sequence and thereby more comprehensively integrate information along the time dimension. This process can be expressed mathematically as follows:

$$F_{lstm} = Bi - LSTM(F'_{fusion}, h_{t-1}) \tag{6}$$

where F'_{fusion} represents the features input into the Bi-LSTM, with the shape $(B, S, features)$. Here $features$ denotes the number of features extracted from the slices. F_{lstm} represents the hidden state at time step t, which has the shape $(B, S, hidden\ size)$.

To mitigate potential overfitting during the training process, a dropout operation is applied to the output of the Bi-LSTM with a dropout rate set at 0.4. This procedure helps to enhance the generalization of the model.

Ultimately, the network utilizes a fully connected layer to use the features obtained from the Bi-LSTM module to predict the probability of a patient belonging to one of three types of pneumonia. The mathematical expression for prediction is as follows:

$$\hat{y} = \text{softmax}(W \cdot F_{lstm} + b) \tag{7}$$

where W and b are respectively the weights and biases of the fully connected layer, and $\hat{y}$ represents the output vector of the network for pneumonia type probabilities.

With this structural design, the model is capable not only of processing features from individual CT slices but also of effectively utilizing temporal association information across the entire sequence, thereby enhancing diagnostic accuracy for complex medical conditions.

4. Results and discussion

In this section, the proposed MSSFN model is evaluated on a custom-built dataset. Additionally, extensive comparative experiments are conducted, benchmarking the MSSFN against other CNN+Bi-LSTM models, as well as state-of-the-art 2D and 3D models. An ablation study is also performed on the MSSFN model. To start with, the experimental environment is introduced.

4.1. Experimental environment

4.1.1. Dataset

Lung CT image data were sourced from Fujian Cancer Hospital, collected by 20 physicians from multiple patients, covering three disease categories: CIP, PLC, and RP, with 89, 99, and 125 cases respectively. In Fig. 3, examples of slices from the three types of pneumonia are displayed. CIP typically shows patterns like Cryptogenic Organizing Pneumonia (COP), Ground-Glass Opacity (GGO), and Nonspecific Interstitial Pneumonia (NSIP). RP often presents with unilateral distribution, fewer pulmonary lobes, and smaller lung volumes, with clearer lesion boundaries. PLC appears on CT as beaded thickening along peribronchovascular bundles and interlobular septa, with polygonal lines, lymph node changes, and pleural effusion. The original data were in DICOM format with a pixel resolution of 512×512.

In the preprocessing stage, the data were first categorized by disease type and then split into training and validation sets in an 8:2 ratio, as detailed in Table 1; subsequently, they were converted into PNG format images.

The specific processing steps are illustrated in Fig. 4. The window width was set to 1500 HU and the window level to −600 HU to optimize the view [38]. A series of morphological operations—including binarization, erosion, identification of the largest connected region, dilation, and filling—were applied to remove non-human body areas.

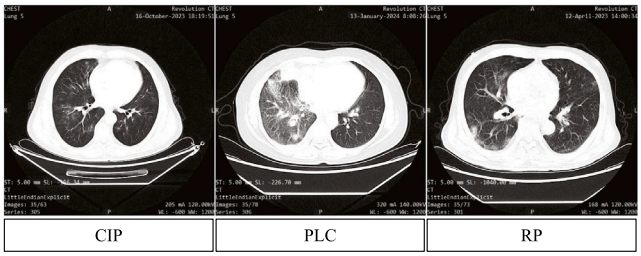


Fig. 3. Examples of CIP, PLC, RP slice.

Table 1
The distribution of self-collected dataset.

	CIP	PLC	RP
Train	71	79	100
Val	18	20	25
Total	89	99	125

Table 2
Hyperparameter settings.

Variables	Values
Batch size	16
Epochs	50
Initial learning rate	5e−5
Optimizer	Adam
Weight decay	1e−5
Scheduler	CosineAnnealingLR
Loss function	CrossEntropyLoss

Additionally, 10 non-essential slices from the beginning and end of each series were removed to focus on the identification of lesion areas. We set a sequence threshold of 50 slices: if a CT series exceeded 50 slices, only the first 50 were retained; if fewer than 50 slices were available, middle slices were repeated until the total reached 50, ensuring consistency in input. Finally, each CT series, after cropping and normalization, was resized to 128 × 128 pixels for input into the network.

4.1.2. Experimental settings and metrics

The hyperparameter settings for this experiment are detailed in Table 2. Given the limited data volume used in this experiment, implementing measures to prevent overfitting was essential [39]. During the training process, L2 regularization was added to the Adam optimizer to penalize the square of the weight parameters, encouraging minimally large weights. In conjunction with a cosine annealing scheduler, which gradually decreases the learning rate according to a cosine function, the model can more finely explore the weight space during training. This helps to prevent the model from settling at local minima, thereby enhancing performance.

The metrics used in the experiment include *Accuracy*, *Precision*, *Recall* and $F1$ *score*, defined in Eqs. (8)–(11).

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{8}$$

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

$$F1\ score = 2 \times \frac{Recall \times Precision}{Recall + Precision} \tag{11}$$

where TP denotes the number of instances correctly predicted as positive; TN denotes the number of instances correctly predicted as negative; FP represents the number of instances incorrectly predicted as positive; FN represents the number of instances incorrectly predicted as negative. The $F1$ *score* is the harmonic mean of recall and precision.

Table 3
Performance of MSSFN.

Classes	Metrics				
	AUC	*Precision*	*Recall*	*F1 score*	*Accuracy*
CIP	0.8494	0.8125	0.7222	0.7647	
PLC	0.9186	0.8421	0.8000	0.8205	0.8254
RP	0.9158	0.8214	0.9200	0.8679	

Additionally, the experimental results utilize Receiver Operating Characteristic (ROC) curve curves and confusion matrices for display. The ROC illustrates the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR) at various threshold settings, serving as a crucial tool for evaluating overall model performance. Confusion matrices provide an intuitive way to understand the classification accuracy across categories, with diagonal values representing the number of instances correctly predicted for each category, while off-diagonal elements indicate misclassifications.

All experiments are conducted using the deep learning framework Pytorch, on a Windows 10 operating system, with an NVIDIA TITAN RTX single GPU.

4.2. Performance of MSSFN

The MSSFN model was employed to conduct a comprehensive assessment of three types of pneumonia diseases, namely CIP, PLC, and RP, which share highly similar characteristics. Detailed data from the evaluations are displayed in Table 3. By analyzing this data, it was found that the MSSFN model achieved an overall classification accuracy of 0.8254, which significantly demonstrates the effectiveness of the model in distinguishing between these three types of pneumonia diseases. Further analysis revealed that PLC demonstrated the most notable precision, scoring at 0.8421. This may be attributed to PLC, which results from the spread of malignant tumor cells, having distinct pathological characteristics from those caused by treatment-related CIP and RP, thereby enabling more accurate recognition of PLC during classification. Moreover, RP exhibited the best performance in terms of recall and F1 scores, reaching 0.9200 and 0.8679 respectively. This indicates that not only is the model capable of accurately identifying this disease, but it also has a high diagnostic precision.

Additionally, to further verify the performance of the MSSFN model, we conducted an in-depth analysis through confusion matrices and ROC curves, as shown in Figs. 5–6. The results from the confusion matrix reveal that there is a certain probability of the CIP category being mistakenly classified as the RP category. This misclassification phenomenon is primarily attributed to the fact that both CIP and RP are associated with inflammatory responses following lung cancer treatment, exhibiting similar pathological characteristics in CT images, which increases the difficulty of distinguishing between these two diseases in the model.

To more comprehensively evaluate the diagnostic capabilities of our network, we plotted the ROC curve and calculated the AUC values for each category. The results show that the AUC values for the three types of pneumonia are 0.8494, 0.9186, and 0.9158 respectively, all of which are close to the ideal state, indicating that the MSSFN model has high accuracy and specificity in diagnosing these diseases.

4.3. Comparisons with other advanced models

4.3.1. Comparisons with other CNN+Bi-LSTM

To validate the effectiveness of multi-scale feature extraction techniques in distinguishing medical images with similar lesions, comparative experiments were conducted using various CNN architectures in

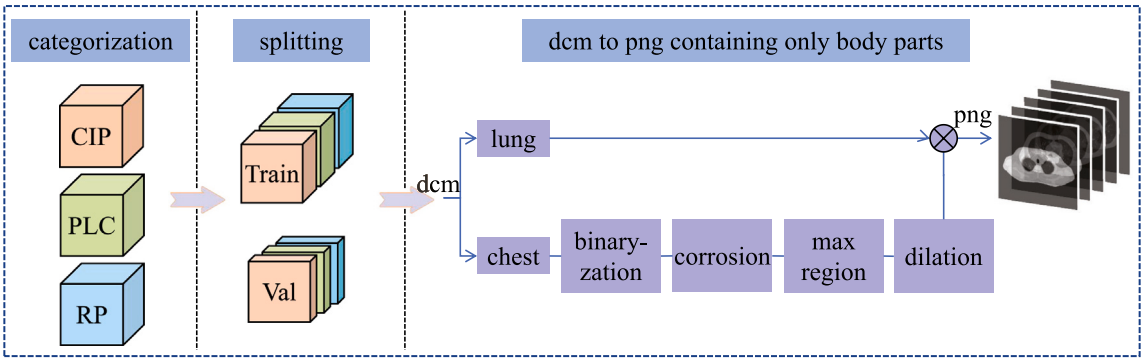


Fig. 4. The preprocessing of self-collected data.

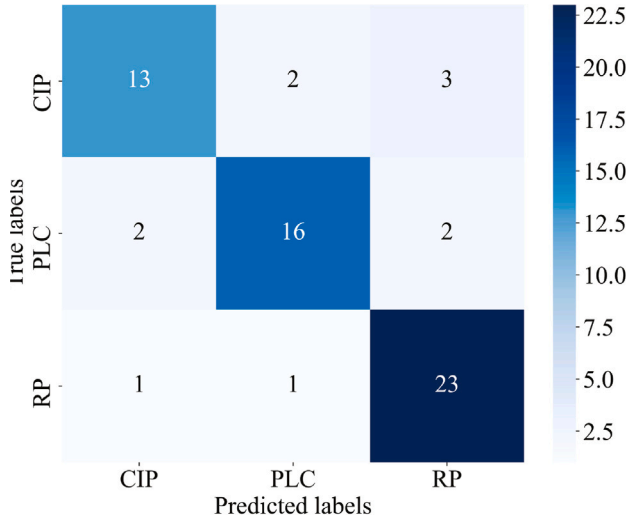


Fig. 5. Confusion Matrix of the MSSFN.

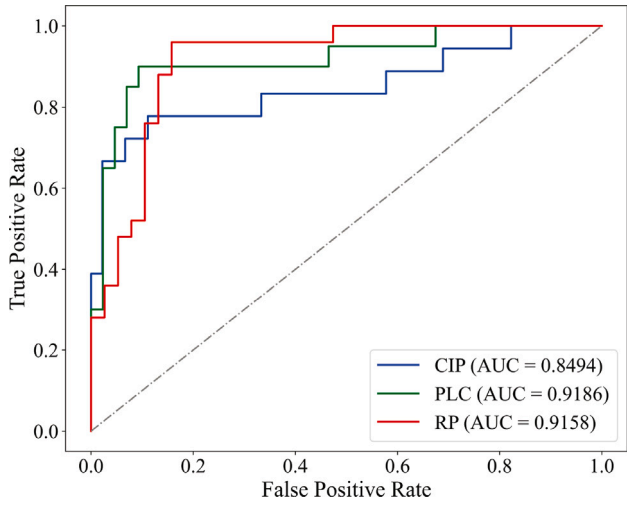


Fig. 6. ROC curves of three pulmonary complications.

this section. To ensure a fair comparison, all networks employed a CNN+Bi-LSTM structure, with the CNN component using parameters pretrained on ImageNet, and the number of features and layers in the Bi-LSTM section being consistent.

Given the limited amount of data, networks with fewer layers such as ShuffleNet [40], EfficientNet [41], ResNet [42], Inception [43], AlexNet [44], and VGG [45] were chosen for experimentation. The detailed results are presented in Fig. 7.

ShuffleNet and EfficientNet structures have significantly fewer parameters compared to other models, but their capability to express features is limited when processing complex CT images, resulting in relatively lower performance. AlexNet uses an 11 × 11 convolution kernel to increase its receptive field, however it performs poorly in terms of Accuracy and F1 scores, which may be due to its inability to effectively capture sufficient local detail information. VGG and ResNet networks increase network depth either by simplifying the network structure or by using residual connections. However, compared to the similarly layered GoogLeNet, ResNet18 and VGG19 have 3% and 8% lower accuracy, respectively.

Compared to the aforementioned networks, the Inception architecture, including GoogLeNet and InceptionV3, significantly improves performance through its innovative multi-scale convolution modules. The Inception module allows for the parallel processing of different sizes of convolution kernels within the same layer, effectively capturing different scales of information in CT images and achieving a good balance between detailed and global features. Leveraging the advantages of multi-scale processing of CT data, the MSSFN model makes full use of Inception modules and has designed the MSFM. The Data show improvements across four metrics, especially in accuracy, which reaches 0.8254, and the recall of RP reaches 0.9200. Therefore, the design of the MSFM demonstrates superiority in handling complex and variable medical images, and the proposed MSSFN model can effectively distinguish between CIP, PLC, and RP.

4.3.2. Comparisons with 2D/3D network

In this section, we showcase the performance of various networks on challenging medical image classification tasks by comparing different advanced 2D and 3D experimental approaches. For fairness, the epoch of all comparison experiments was set to 50, the batch size was set to 16, and all of them used the Adam optimizer with CosineAnnealingLR to adjust the learning rate, with the initial learning rate set to 5e−5. These networks include the redesigned COVID-Net [46], SE-ResNet50 [47], DeCoVNet [20], and Med3D [48]. The redesigned COVID-Net and SE-ResNet50 utilize 2D approaches, while DeCoVNet and Med3D employ 3D methods.

The redesigned COVID-Net is an enhancement of the original COVID-Net architecture, incorporating a dual-branch structure to augment feature extraction capabilities and introducing batch normalization layers to stabilize the learning process and better adapt to statistical variations across different sites. SE-ResNet50 calculates the probability of each slice belonging to a specific type and then classifies based on the average probability across all slices. DeCoVNet is a three-dimensional network that includes a 3D convolutional layer, two 3D residual modules, and a classifier. Med3D, based on the ResNet architecture, is a deep 3D convolutional network specifically designed for processing and analyzing 3D medical imaging data.

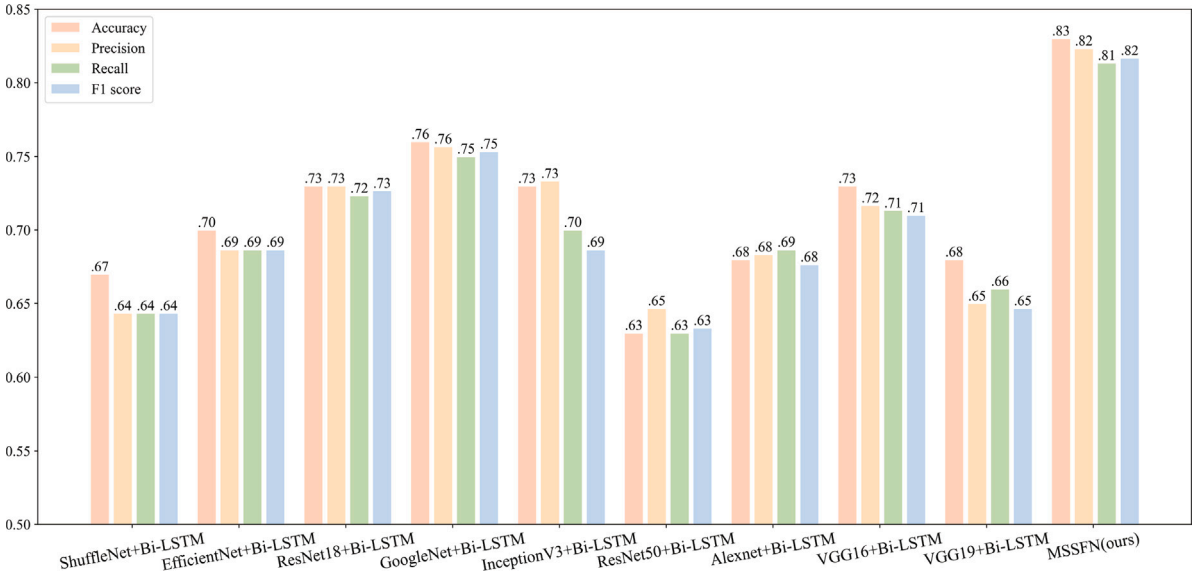


Fig. 7. Performance comparison of MSSFN and other CNN+Bi-LSTM models.

Table 4
Performance comparison of state-of-the-art 2D/3D CNNs.

Model	Classes	Metrics			
		Precision	*Recall*	*F1 score*	*Accuracy*
redesigned COVID-Net [46]	CIP	0.6723	0.4415	0.5330	0.6205
	PLC	0.6127	0.6223	0.6175	
	RP	0.6125	0.7618	0.6790	
SE-ResNet50 [47]	CIP	0.2828	1.0000	0.4409	0.6869
	PLC	0.7523	0.6029	0.6694	
	RP	0.8413	0.6420	0.7283	
DeCoVNet [20]	CIP	0.5618	0.5022	0.5303	0.6169
	PLC	0.6875	0.6537	0.6702	
	RP	0.6446	0.6519	0.6482	
Med3D [48]	CIP	0.6943	0.5042	0.5842	0.6384
	PLC	0.6066	0.6028	0.6047	
	RP	0.6344	0.6079	0.6209	
MSSFN (**Ours**)	CIP	0.8125	0.7222	0.7647	0.8254
	PLC	0.8421	0.8000	0.8205	
	RP	0.8214	0.9200	0.8679	

Table 5
Ablation studies of MSFM and Bi-LSTM.

Model	Classes	Metrics			
		Precision	*Recall*	*F1 score*	*Accuracy*
GoogLeNet + Bi-LSTM	CIP	0.6712	0.5623	0.6101	0.7117
	PLC	0.7418	0.7011	0.7194	
	RP	0.7216	0.8412	0.7754	
GoogLeNet + MSFM	CIP	0.5714	0.4421	0.4966	0.7012
	PLC	0.6710	0.8018	0.7293	
	RP	0.8006	0.8016	0.8000	
MSSFN (**Ours**)	CIP	0.8125	0.7222	0.7647	0.8254
	PLC	0.8421	0.8000	0.8205	
	RP	0.8214	0.9200	0.8679	

It is noteworthy that an evaluation was conducted on two types of 3D CNN architectures, DeCoVNet and Med3D, for classification tasks, even though these incorporate steps related to image segmentation. As shown in Table 4, experimental results indicate that accuracies achieved by these networks were 0.6205, 0.6869, 0.6169, and 0.6384 respectively, which clearly fall short of the performance offered by the pretrained CNN+Bi-LSTM structure. On one hand, the unique situation of having fewer cases but more slices provides the CNN+LSTM structure with an advantage and specificity when processing this dataset. On the other hand, transfer learning plays a crucial role in this study. Utilizing parameters that were pretrained on large datasets to initialize the network allows for a significant enhancement of network performance. This approach not only accelerates the convergence speed of the model but also significantly improves key metrics such as accuracy and recall.

4.4. Ablation study

In this section, to further validate the effectiveness of the core components of our model, a series of ablation experiments was designed. The results of these experiments are shown in Table 5.

Experimental results show that the proposed MSSFN model exhibits noticeable improvements across all metrics. The introduction of the Multi-Scale Feature Fusion Module (MSFM) leads to a significant improvement of the model accuracy by 11.37%. This indicates that the MSFM effectively integrates features from different scales, thereby enhancing the multidimensional perceptive ability of the model. Lesion information in CT images manifests differently at various spatial scales. Through multi-scale processing, MSFM not only captures fine local features, but also identifies structural changes over larger regions. As a result, MSFM provides the model with richer and more comprehensive feature representations, improving its expressiveness and robustness in dealing with complex lesions. On the other hand, the removal of the Bi-LSTM module results in a 12.42% decrease in accuracy, highlighting the richness of spatial information within CT sequences and the critical role of Bi-LSTM in capturing temporal dynamics. CT images contain not only static structural information but also temporal features related to lesion progression. The Bi-LSTM module utilizes contextual information from both preceding and succeeding time steps to capture the dynamic features of the CT sequence over time.

To visually demonstrate the performance of models in recognizing three types of lung diseases, we detailed the results of each experimental configuration through ROC curves. As shown in Fig. 8, MSSFN exhibited the most outstanding performance, with an AUC value reaching 0.9012, while GoogLeNet+Bi-LSTM and GoogLeNet+MSFM achieved an AUC of 0.8890 and 0.8862. Particularly in areas with low false positive rates, MSSFN demonstrated a higher true positive rate, highlighting its diagnostic accuracy advantages. These results not only verify the effectiveness of each model component but also emphasize their potential application in improving the diagnosis and treatment of critically ill patients.

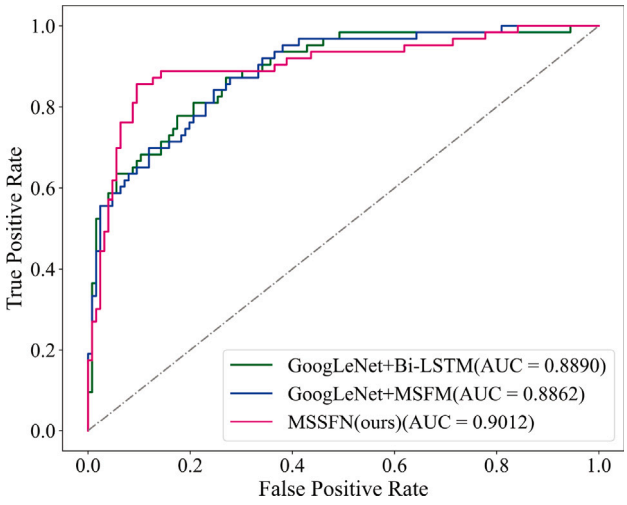


Fig. 8. Aggregate ROC curves of multiple models on three pulmonary complications.

5. Conclusion

In this paper, a novel deep learning based diagnosis method, MSSFN, is proposed to accurately distinguish three types of diseases from chest CT images. Firstly, a lung CT dataset including three diseases, CIP, PLC, and RP, was established and meticulously preprocessed to ensure the consistency and standardization of the input data. Subsequently, the MSSFN model proposed in this paper leverages the pre-trained GoogLeNet front 17-layer network to extract initial shallow features. The model then utilizes a multi-scale feature extraction module to learn the high-dimensional features of the lesion region. Eventually, the model employs a spatial information learning technique to classify the aforementioned three lung diseases. This approach enhances the multidimensional feature representation of medical images, optimizing the generalization and computational efficiency of the model.

In addition, by comparing with a variety of CNN+Bi-LSTM models as well as current state-of-the-art 2D and 3D models, MSSFN shows advantages in feature extraction and spatial information learning, especially when dealing with medical images with complex lesions, achieving an accuracy of 82.54% on three lung disease classification problems. The ablation experiments further confirmed the key role of the multiscale fusion module and Bi-LSTM in improving model performance, highlighting the importance of these components in integrating features at different scales and enhancing the spatial information processing capability of CT sequence data.

While MSSFN achieves satisfactory results in lung disease classification, several aspects still need to be improved. On the one hand, its adaptability to variations in imaging protocols and patient-specific factors remains insufficiently explored, impacting its generalization in clinical practice. On the other hand, the current feature extraction methods may not fully capture complex or subtle variations in CT images, which could potentially affect performance. Future work will concentrate on two main directions: firstly, the extension of MSSFN to other CT medical image analysis tasks to ascertain its applicability in diverse scenarios [49,50]. Secondly, by introducing advanced feature extraction techniques [51,52] to further refine the structure of MSSFN, aiming to achieve higher diagnostic accuracy and stronger adaptability, thus providing clinicians with more precise and reliable diagnostic support tools.

CRediT authorship contribution statement

Hongfu Zeng: Writing – original draft, Methodology, Investigation. **Xinyu Li:** Writing – original draft, Methodology, Investigation. **Haipeng Xu:** Validation, Resources, Methodology, Conceptualization. **Keyi Yu:** Writing – review & editing, Methodology, Investigation. **Huihua Hu:** Writing – review & editing, Validation, Investigation. **Peishu Wu:** Writing – review & editing, Visualization, Validation, Investigation. **Nianyin Zeng:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology.

Declarations

The study was approved by the Ethics Committee of Fujian Cancer Hospital (approval no. K2024-433-01) and was conducted in accordance with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Informed consent was waived by our Institutional Review Board because of the retrospective nature of our study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Y. Park, Y. Im, J. Kwon, J. Han, M. Ahn, J. Kim, S. Um, Y. Choi, Prognostic value of pretherapeutic FDG PET/CT in non-small cell lung cancer with pulmonary lymphangitic carcinomatosis, Sci. Rep. 13 (1) (2023).
[2] L. Shi, J. Yang, H. Yao, Multiple regression analysis of risk factors related to radiation pneumonitis, World J. Clin. Cases 11 (5) (2023) 1040–1048.
[3] M. Lin, D. Zang, C. Liu, X. Han, J. Chen, Immune checkpoint inhibitor-related pneumonitis: Research advances in prediction and management, Front. Immunol. 15 (2024).
[4] J. Cheng, Y. Pan, W. Huang, K. Huang, Y. Cui, W. Hong, L. Wang, D. Ni, P. Tan, Differentiation between immune checkpoint inhibitor-related and radiation pneumonitis in lung cancer by CT radiomics and machine learning, Med. Phys. 49 (3) (2022) 1547–1558.
[5] X. Lu, J. Wang, T. Zhang, Z. Zhou, L. Deng, X. Wang, W. Wang, W. Liu, W. Tang, Z. Wang, J. Wang, W. Jiang, N. Bi, L. Wang, Comprehensive pneumonitis profile of thoracic radiotherapy followed by immune checkpoint inhibitor and risk factors for radiation recall pneumonitis in lung cancer, Front. Immunol. 13 (2022).
[6] Q. Qiu, L. Xing, Y. Wang, A. Feng, Q. Wen, Development and validation of a radiomics nomogram using computed tomography for differentiating immune checkpoint inhibitor-related pneumonitis from radiation pneumonitis for patients with non-small cell lung cancer, Front. Immunol. 13 (2022).
[7] W. Ehab, L. Huang, Y. Li, Unet and variants for medical image segmentation, Int. J. Netw. Dyn. Intell. 3 (2) (2024) 100009.
[8] H. Chen, R. Wu, C. Tao, W. Xu, H. Liu, C. Xu, M. Jian, Multi-scale class attention network for diabetes retinopathy grading, Int. J. Netw. Dyn. Intell. 3 (2) (2024) 100012.
[9] X. Li, M. Li, P. Yan, G. Li, Y. Jiang, H. Luo, S. Yin, Deep learning attention mechanism in medical image analysis: Basics and beyonds, Int. J. Netw. Dyn. Intell. 2 (1) (2023) 93–116.
[10] Y. Pathak, A. Tiwari, S. Stalin, S. Singh, K. Shukla, Deep transfer learning based classification model for COVID-19 disease, Innov. Res. Biomed. Eng. 43 (2) (2022) 87–92.
[11] S. Wang, L. Fernandes, Z. Zhu, Y. Zhang, AVNC: Attention-based VGG-style network for COVID-19 diagnosis by CBAM, IEEE Sens. J. 22 (18) (2022) 17431–17438.
[12] Q. Zhang, J. Shu, C. Chen, Z. Teng, Z. Gu, F. Li, J. Kan, Optimization of pneumonia CT classification model using RepVGG and spatial attention features, Front. Med. 10 (2023).
[13] W. Fang, B. Shen, A. Pan, L. Zou, B. Song, A cooperative stochastic configuration network based on differential evolutionary sparrow search algorithm for prediction, Syst. Sci. Control. Eng. 12 (1) (2024) 2314481.
[14] M. Abdar, A. Fahami, L. Rundo, P. Radeva, F. Frangi, R. Acharya, A. Khosravi, H. Lam, A. Jung, S. Nahavandi, Hercules: Deep hierarchical attentive multilevel fusion model with uncertainty quantification for medical image classification, IEEE Trans. Ind. Inform. 19 (1) (2023) 274–285.

[15] J. Yin, F. Yan, Y. Liu, G. He, Y. Zhuang, An overview of simultaneous localisation and mapping: towards multi-sensor fusion, Int. J. Syst. Sci. 55 (3) (2024) 550–568.
[16] O. Tas, S. Yavuz, Enhancing lung cancer survival prediction: 3D CNN analysis of CT images using novel GTV1-SliceNum feature and PEN-BCE loss function, Diagnostics 14 (12) (2024).
[17] M. Solar, P. Aguirre, Deep learning techniques to process 3D chest CT, J. Univers. Comput. Sci. 30 (6) (2024) 758–778.
[18] M. Fallahpoor, S. Chakraborty, T. Heshejin, H. Chegeni, J. Horry, B. Pradhan, Generalizability assessment of COVID-19 3D CT data for deep learning-based disease detection, Comput. Biol. Med. 145 (2022).
[19] K. Saoudi, K. Bdirina, K. Guesmi, Robust estimation and control of uncertain affine nonlinear systems using predictive sliding mode control and sliding mode observer, Int. J. Syst. Sci. 55 (7) (2024) 1480–1492.
[20] X. Wang, X. Deng, Q. Fu, Q. Zhou, J. Feng, H. Ma, W. Liu, C. Zheng, A weakly-supervised framework for COVID-19 classification and lesion localization from chest CT, IEEE Trans. Med. Imaging 39 (8) (2020) 2615–2625.
[21] M. Sheng, W. Ding, W. Sheng, Differential evolution with adaptive niching and reinitialisation for nonlinear equation systems, Int. J. Syst. Sci. 55 (10) (2024) 2172–2186.
[22] N. Sindhura, M. Pai, N. Bhat, M. Pai, A review of deep learning and generative adversarial networks applications in medical image analysis, Multimedia Syst. 30 (3) (2024).
[23] G. Latif, J. Alghazo, A. Khan, G. Ben K. Fawagreh, N. Mohammad, Deep convolutional neural network (CNN) model optimization techniques-review for medical imaging, Aims Math. 9 (8) (2024) 20539–20571.
[24] N. Dzhaynakbaev, N. Kurmanbekkyzy, A. Baimakhanova, I. Mussatayeva, 2D-CNN architecture for accurate classification of COVID-19 related pneumonia on X-ray images, Int. J. Adv. Comput. Sci. Appl. 15 (1) (2024) 905–917.
[25] R. Mastouri, N. Khlifa, H. Neji, S. Hantous-Zannad, Performance analysis of various deep learning models based on max-min CNN for lung nodule classification on CT images, Mach. Vis. Appl. 35 (4) (2024).
[26] X. Fan, X. Feng, Y. Dong, H. Hou, COVID-19 CT image recognition algorithm based on transformer and CNN, Displays 72 (2022).
[27] Y. Xie, Y. Xia, J. Zhang, Y. Song, D. Feng, M. Fulham, W. Cai, Knowledge-based collaborative deep learning for benign-malignant lung nodule classification on chest CT, IEEE Trans. Med. Imaging 38 (4) (2019) 991–1004.
[28] E. Akbacak, N. Muzoglu, An efficient and robust 3D medical image classification approach based on 3D CNN, time-distributed 2D CNN-BLSTM models, and mRMR feature selection, Comput. Intell. 40 (5) (2024).
[29] K. Abinaya, B. Sivakumar, A deep learning-based approach for cervical cancer classification using 3D CNN and vision transformer, J. Imaging Inform. Med. 37 (1) (2024) 280–296.
[30] F. Liao, M. Liang, Z. Li, X. Hu, S. Song, Evaluate the malignancy of pulmonary nodules using the 3-D deep leaky noisy-OR network, IEEE Trans. Neural Netw. Learn. Syst. 30 (11) (2019) 3484–3495.
[31] H. Huang, R. Wu, Y. Li, C. Peng, Self-supervised transfer learning based on domain adaptation for benign-malignant lung nodule classification on thoracic CT, IEEE J. Biomed. Heal. Inform. 26 (8) (2022) 3860–3871.
[32] H. Naeem, A. Bin-Salem, A CNN-LSTM network with multi-level feature extraction-based approach for automated detection of coronavirus from CT scan and X-ray images, Appl. Soft Comput. 113 (A) (2021).
[33] M. Srikantamurthy, S. Rallabandi, B. Dudekula, S. Natarajan, J. Park, Classification of benign and malignant subtypes of breast cancer histopathology imaging using hybrid CNN-LSTM based transfer learning, BMC Med. Imaging 23 (1) (2023).
[34] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, A. Rabinovich, Going deeper with convolutions, in: IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2015, pp. 1–9.
[35] R. Khattab, R. Abdelmaksoud, S. Abdelrazek, Automated detection of COVID-19 and pneumonia diseases using data mining and transfer learning algorithms with focal loss from chest X-ray images, Appl. Soft Comput. 162 (2024).
[36] G. Liang, L. Zheng, A transfer learning method with deep residual network for pediatric pneumonia diagnosis, Comput. Methods Programs Biomed. 187 (2020).
[37] N. Patel, R. Murugan, G. Srivastava, R. Maddikunta, G. Yenduri, R. Gadekallu, R. Chengoden, An explainable transfer learning framework for multi-classification of lung diseases in chest X-rays, Alex. Eng. J. 98 (2024) 328–343.
[38] L. Li, Y. Hua, Y. Liu, F. Huang, Study on fast fractal image compression algorithm based on centroid radius, Syst. Sci. Control. Eng. 12 (1) (2024) 2269183.
[39] R. Wang, J. Liang, The effect of multiscale parameters on the spiking properties of the morphological neuron with excitatory autapse, Syst. Sci. Control. Eng. 12 (1) (2024) 2313865.
[40] X. Zhang, X. Zhou, M. Lin, R. Sun, ShuffleNet: An extremely efficient convolutional neural network for mobile devices, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2018, pp. 6848–6856.
[41] M. Tan, V. Le, EfficientNet: Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, Vol. 97, 2019.
[42] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2016, pp. 770–778.
[43] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: IEEE Conference on Computer Vision and Pattern Recognition, CVPR, 2016, pp. 2818–2826.
[44] A. Krizhevsky, I. Sutskever, E. Hinton, ImageNet classification with deep convolutional neural networks, Commun. ACM 60 (6) (2017) 84–90.
[45] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, Comput. Sci. (2014).
[46] Z. Wang, Q. Liu, Q. Dou, Contrastive cross-site learning with redesigned net for COVID-19 CT classification, IEEE J. Biomed. Heal. Inform. 24 (10) (2020) 2806–2813.
[47] F. Zheng, L. Li, M. Zhang, Y. Song, Z. Huang, Y. Chong, Z. Chen, H. Zhu, J. Wu, W. Chen, Y. Lu, Y. Yang, Y. Zha, H. Zhao, J. Shen, Accurately discriminating COVID-19 from viral and bacterial pneumonia according to CT images via deep learning, Interdiscip. Sci.- Comput. Life Sci. 13 (2) (2021) 273–285.
[48] S. Chen, K. Ma, Y. Zheng, Med3D: Transfer learning for 3D medical image analysis, 2019, arXiv, abs/1904.00625.
[49] D. Lteif, S. Sreerama, A. Bargal, A. Plummer, R. Au, B. Kolachalama, Disease-driven domain generalization for neuroimaging-based assessment of Alzheimer's disease, Hum. Brain Mapp. 45 (8) (2024).
[50] P. Wu, Z. Wang, B. Zheng, H. Li, E. Alsaadi, N. Zeng, AGGN: Attention-based glioma grading network with multi-scale feature extraction and multi-modal information fusion, Comput. Biol. Med. 152 (2023).
[51] Q. Zhou, Z. Huang, M. Ding, X. Zhang, Medical image classification using light-weight CNN with spiking cortical model based attention module, IEEE J. Biomed. Heal. Inform. 27 (4) (2023) 1991–2002.
[52] H. Zhu, J. Wang, S. Wang, R. Raman, M. Gorriz, Y. Zhang, An evolutionary attention-based network for medical image classification, Int. J. Neural Syst. 33 (3) (2023).



Hongfu Zeng received a Bachelor's Degree in Clinical Medicine from Fujian Medical University in 2005. He graduated with a Master's Degree in Oncology from the same institution in 2014. Currently, he serves as a Physician in the Department of Thoracic Oncology at Fujian Cancer Hospital, Fuzhou, China. His professional expertise lies in the comprehensive management of lung cancer and interventional pulmonology.



Xinyu Li received the bachelor degree in Automation from Qingdao University of Science and Technology, China, in 2023. She is currently pursuing a master degree in Control Engineering at Xiamen University, China. Her research interests include computer vision and medical image processing.



Haipeng Xu earned his Bachelor's Degree from Zhengzhou University, Master's Degree from Sun Yat-sen University, and Ph.D. from Tianjin Medical University. Since 2006, he has been working at Fujian Cancer Hospital, where he is primarily engaged in the management of thoracic malignancies, including lung cancer, esophageal cancer, and thymic malignancies. He has participated in numerous international and domestic clinical research trials.



Keyi Yu received B.S. degree in management from Qingdao University of technology, Qingdao, China, in 2019. She is currently a graduate student with the National Institute for Data Science in Health and Medicine, School of Medicine, Xiamen University, Xiamen, China. Her current research interests include gene expression analysis, advanced deep learning algorithms and their clinical applications.



Huihua Hu received a Bachelor's Degree in Clinical Medicine from Fujian Medical University in 2005. Currently, she serves as a Physician in the Department of Thoracic Oncology at Fujian Cancer Hospital, Fuzhou, China.



Peishu Wu received the bachelor degree in Measurement & Control Technology and Instrumentation from Tianjin University of Science and Technology, Tianjin, China, in 2020. He is currently pursuing the Ph.D. degree in Instrument Science and Technology at Xiamen University, Xiamen, China. His research interests include computer vision and industrial defect detection techniques.



Nianyin Zeng was born in Fujian Province, China, in 1986. He received the B.Eng. degree in electrical engineering and automation in 2008 and the Ph. D. degree in electrical engineering in 2013, both from Fuzhou University. From October 2012 to March 2013, he was a RA in the Department of Electrical and Electronic Engineering, the University of Hong Kong. From September 2017 to August 2018, he as an ISEF Fellow founded by the Korea Foundation for Advance Studies and also a Visiting Professor at the Korea Advance Institute of Science and Technology.

Currently, he is a Professor with the Department of Instrumental & Electrical Engineering of Xiamen University. His current research interests include intelligent data analysis, computational intelligent, time-series modeling and applications. He is the author or co-author of several technical papers and also a very active reviewer for many international journals and conferences.

Prof. Zeng is currently serving as Associate Editors for Neurocomputing, Evolutionary Intelligence, and Frontiers in Medical Technology, and also Editorial Board members for Computers in Biology and Medicine, Biomedical Engineering Online, and Mathematical Problems in Engineering.